

Inner Loop: Query Optimization

Inner-Loop Rounds

Kept docs: mean pred.



Round 1 broad SABRE heuristics

“SABRE” adaptive lookahead decay swap scoring weighted ...

Structured Scaling ...	6,968	✓
dSABRE: A SABRE-...	6,962	✓
...		

Knowledge State

... suggest weighting lookahead gates by dependency depth using an exponential factor ...

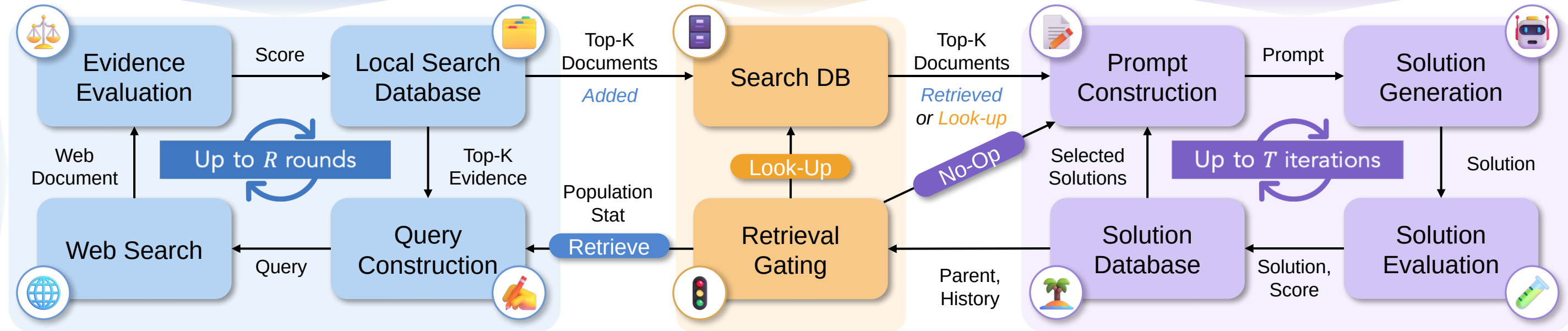
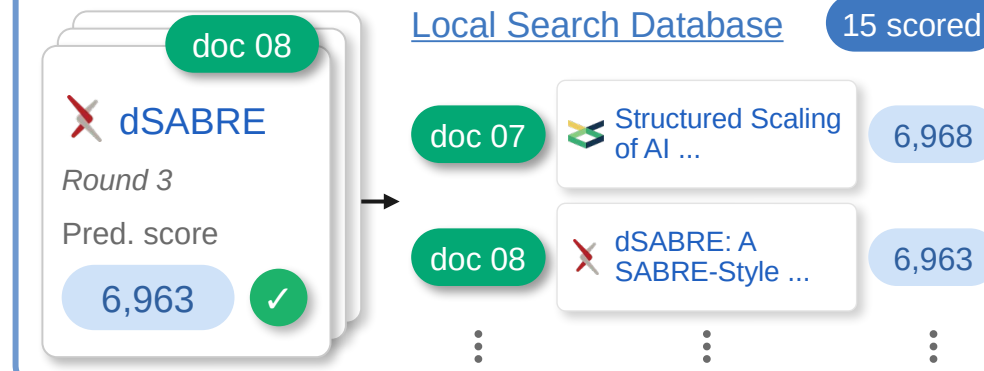
Round 3 depth-weighted lookahead

“precomputed_extended_set_logical_pairs” dependency depth BFS ...

dSABRE: A SABRE-...	6,963	✓
LightSABRE: A ...	6,959	
...		

Knowledge State

... depth-weighted lookahead is the most promising next modification ...



Query Construction

Round 3

Knowledge State after R2

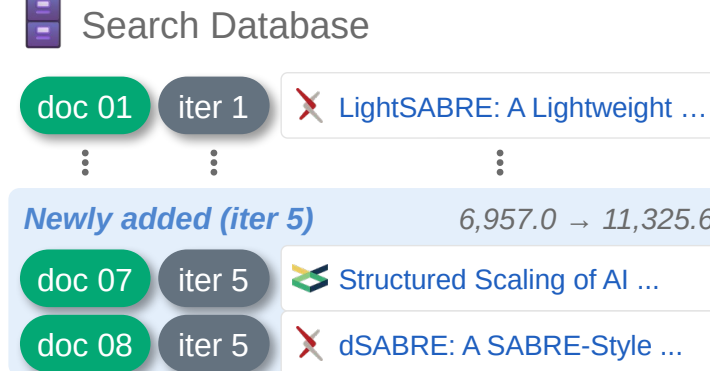
... do not provide actionable repository-specific routing information ... dependency-depth weighting hypothesis ... remains plausible ...



Assign dependency depths to extended-set gates

"precomputed_extended_set_logical_pairs" dependency depth BFS layer weighted lookahead Rust SABRE

Knowledge-Gap Retrieval Gating



Iteration 5

gate decision

Knowledge State

Stored docs give SABRE/LightSABRE principles but no repository-specific Rust API ...

Reasoning

Additional evidence would help ... current search documents are largely conceptual ...

Retrieve

Look-Up

No-Op

Outer Loop: Solution Optimization

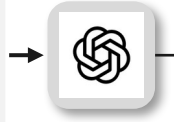
Prompt

[Task] Rust qubit routing

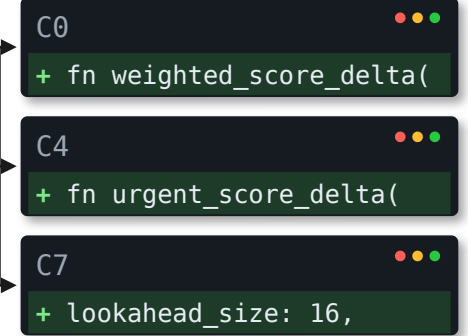
[Parent] 690e7bf1

[History] 3 attempts

[Docs] 07 08



× 8 in parallel



Population Database

6 solutions

id	Solution	Eval. Score
c302e9b5	[Diagram]	11,325.6
690e7bf1	[Diagram]	6,957.0
ee6d0860	[Diagram]	6,957.0
...	...	...

best of 8 valid children is kept